

CLINICAL PRACTICE GUIDELINE

World Association for Bronchology and Interventional Pulmonology (WABIP) Guidelines on Airway Stenting for Benign Central Airway Obstruction

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ABSTRACT

Benign central airway obstruction (BCAO) related strictures include post-intubation tracheal stenosis, post-tracheostomy tracheal stenosis, post-surgical or anastomotic stenosis, stenosis related to inflammatory disorders or infectious disorders, chemical injury or post radiotherapy. Techniques and thresholds for airway stenting for each of these unique conditions vary significantly across centres due to lack of guidance. The management of the airway stents once placed due to scarcity of data also has a similar knowledge void. This document aims to address these knowledge gaps by addressing relevant questions related to airway stenting in BCAO. A multidisciplinary, international group of 20 experts from 19 institutions across 12 countries with extensive experience in airway stenting in BCAO was convened as part of this guideline document through the World Association of Bronchology and Interventional Pulmonology (WABIP). A literature search was performed for reports addressing eight clinically relevant PICO questions. Recommendations addressing these questions were formulated using a systematic process involving

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1 | Introduction

Benign central airway obstruction (BCAO) refers to the narrowing of the central airways (trachea, main stem bronchi or the bronchus intermedius) by histologically non-malignant processes. Mechanistically, BCAO is due to strictures (with or without malacia), pure dynamic forms of obstruction (tracheo-bronchomalacia (TBM) and excessive dynamic airway collapse (EDAC)) as well as exophytic endoluminal lesions (e.g., respiratory papillomatosis, schwannoma, granular tumour, granulation tissue) or due to pure extrinsic compression (e.g., goitre, vascular abnormalities). While histologically these entities are benign, their clinical presentation can often cause respiratory failure and death due to critical airway obstruction.

Benign strictures include post-intubation tracheal stenosis (PITS), post-tracheostomy tracheal stenosis (PTTS), post-surgical or anastomotic stenosis, stenosis related to inflammatory disorders (e.g., granulomatosis with polyangiitis (GPA)), infectious disorders (e.g., tuberculosis), chemical injury (such as toxic inhalation or reflux) or post radiation therapy. Physiologic, bronchoscopic and computed tomographic assessment of the extent, location, morphology and severity of airway narrowing are essential for optimal management, specifically for avoiding complications and offering definitive therapy when feasible. Several attempts to classify BCAO have been made over the years [1, 2]; however, these have not been widely adopted due to the heterogeneous nature of BCAO [3, 4]. While subjective assessments of severity, even by experienced bronchoscopists, are frequently inaccurate [5, 6], objective assessments of the severity of airway narrowing and symptoms have been demonstrated to be feasible [7–9]. Accurate evaluation of airway narrowing could help direct interventions towards those whose symptoms are truly attributable to the stenosis and not due to underlying comorbidities. Strictures cause turbulent airflow, increasing airway resistance. In fact, airway pressure drop across the narrowing correlates with flow limitation and is noted to significantly increase only when stenosis exceeds 70%, but extent and morphology of the stricture (concentric vs. elliptical vs. triangular) also impact flow limitation [9].

The goal of any intervention in a patient with BCAO, whether medical, endoscopic, or open surgical, is to relieve symptoms of dyspnoea or difficulty clearing secretions, improve quality of life, and mitigate the risk of recurrence requiring subsequent interventions [10]. Simple stenoses are defined as <1 cm in vertical extent and without associated malacia [11]. An example of a simple stenosis is the short, web-like fibrotic stricture without cartilaginous involvement, resulting from endotracheal tube or tracheostomy cuff-related injury, or some autoimmune disorders (e.g., GPA) (Figure 1) Stricture complexity, based on length or airway wall involvement (i.e., malacia or full thickness injury) is a predictor of failure of endoscopic management or need for repeated endoscopic interventions [12]. Complex strictures may also be seen in association with cuff-related injury when full thickness airway wall injury occurs (such as hourglass-like stenosis) (Figure 2), but they are commonly associated with post-tracheostomy stomal stenoses which are usually of pseudo-glottic morphology (A frame; triangular) resulting from abnormal healing of a fractured or injured anterior tracheal ring during tracheostomy tube placement (Figure 3).

Bronchoscopic interventions for stenosis management usually involve radial cuts (using lasers, electrocautery or scissors) and dilation (using controlled radial expansion balloons, a rigid bronchoscope or dilators). Studies have not consistently demonstrated a clear benefit of the use of local steroid injection [13, 14] or mitomycin C application [15, 16] to reduce the stenosis recurrence rates. In select scenarios, airway stenting is considered. There are no randomised, comparative trials of airway stenting in BCAO, and published data is scarce, largely single-centred and retrospective. Consequently, there are no guidelines addressing airway stenting in BCAO. This document, presented as a clinical practice guideline, aims to address this knowledge gap. Our recommendations will address specific clinical entities in each question, with a focus on strictures and dynamic obstruction. However, they may not be extrapolated to BCAO in children. This document will not cover stenosis related to benign endoluminal tumours, extrinsic compression or subglottic stenosis. Subglottic stenoses are largely managed endoscopically or by open surgery and airway stenting in this region is rarely

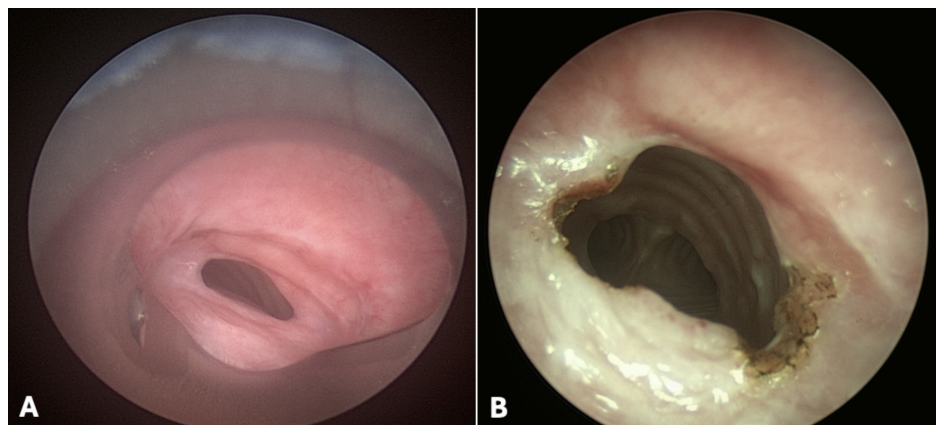


FIGURE 1 | A simple web-like tracheal stenosis seen before (Panel A) and after (Panel B) endoscopic laser radial cuts.

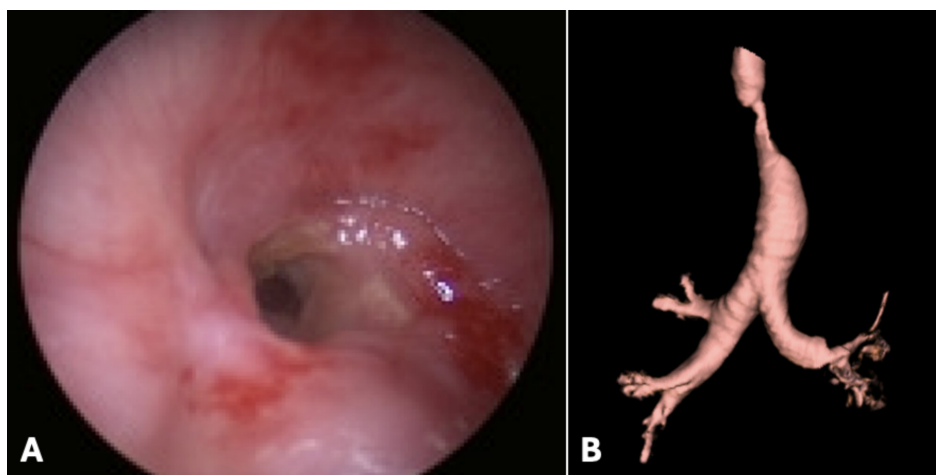


FIGURE 2 | A complex hour-glass stenosis noted during bronchoscopy (A) and a 3D reconstruction of the patient's CT scan (B).



FIGURE 3 | A post-tracheostomy stomal complex tracheal stenosis of pseudo-glottic morphology noted during bronchoscopy.

used in extraordinary circumstances, beyond the purpose of these stenting guidelines.

Airway stenting for BCAA, in general, must only be considered electively in medically inoperable patients or surgically unresectable stenoses, or in emergent scenarios when surgery is not feasible. In such scenarios, airway stenting can be pursued when the stenosis morphology, location or extent are not amenable to endoscopic interventions, when endoscopic failure can be easily predicted (full thickness, complex stenoses; associated malacia) or in those who undergo 'frequent' endoscopic interventions. The definition of 'frequent' is a shared one between the patient and provider and must factor in not only the feasibility for stenting or surgery, but also the patient's preferences. Interventions must be performed meticulously without collateral injury to the adjacent normal airway mucosa. Bleeding and mucosal edema

from any airway manipulation can lead to worsening CAO and excessive thermal damage to the mucosa can lead to recurrence and even extend the degree of stenosis. Airway stents selection must be carefully considered. Oversized stents pose high radial force and may worsen mucosal injury risking full-thickness injuries, vascular perforation or aerodigestive fistulae, and undersized stents can migrate down to the carina or up to the larynx causing a potential critical airway obstruction. While interventions can be performed with flexible or rigid bronchoscopes, silicone stenting can only be performed with the latter. It is noteworthy, that due to increased risks of complications, in 2005 the US Food and Drug Administration recommended against self-expandable metallic stents for BCAA. While this largely holds true for uncovered stents, limited published data may justify the use of contemporary evolved fully covered stents in select scenarios [17].

This guideline document is focused on the benefits and complications of stenting for strictures, TBM, and EDAC. We emphasize that these disorders are managed by experienced operators skilled in these therapeutic procedures and in a facility with advanced airway management expertise. We acknowledge that decisions involving airway stenting involve a complex interplay of patient-and procedure-dependent factors, and extrapolation of our recommendations to a particular patient's unique scenario requires careful evaluation by qualified healthcare providers at the patient's bedside. In some centres, the decisions to place a non-emergent airway stent for BCAA are discussed in a multi-disciplinary meeting involving thoracic surgeons, otolaryngologists, and interventional pulmonologists experienced in managing such patients [18, 19].

2 | Methodology

2.1 | Literature Search

We employed expert opinion to formulate eight clinically relevant questions concerning the use of airway stenting for managing patients with benign central airway obstruction (BCAO). Questions were formulated using the patient, intervention, comparison, and outcome (PICO) format.

An electronic search of Medline via the PubMed interface was conducted using the following keywords: 'stents', 'stenting', 'airway obstruction', 'tracheal stenosis', 'bronchial stenosis', 'laryngotracheal stenosis', 'bronchoscopy', 'benign airway obstruction', 'surveillance', and 'follow-up', combined with Boolean operators to capture studies published between 1985 and February 28, 2024. The search was limited to articles in English, yielding a total of 892 studies.

From these, relevant studies including adult patients were screened based on our predefined questions. We reviewed 240 full texts, including relevant references, and ultimately included 138 articles in our final recommendations (Figure 4). For these articles, we created strength of evidence tables for relevant PICO-based studies when available. Additionally, we assessed the risk of bias using the Newcastle-Ottawa Scale (NOS).

We utilised the GRADE (Grading of Recommendations, Assessment, Development, and Evaluation) methodology to evaluate the strength of the recommendations. For questions lacking corresponding evidence (PICO 6), we incorporated expert consensus to generate a best practice statement. Summary evidence tables for PICO questions are provided in the [Supporting Information Appendix](#).

A core writing group (UC, AA, SM, AR) first drafted distinct recommendations based on the above listed criteria. The document was then circulated amongst all panel members, inviting opinion about existing recommendations and any additional recommendations. The panel included 18 pulmonologists and 2 thoracic surgeons with experience in using airway stenting for B CAO, from 19 different institutions in 12 countries. The document was revised based on suggestions from all members. The revised document was then shared with all panellists and subsequently discussed during a conference call with 70% participation. During the conference call, individual suggestions were reviewed and incorporated in real time. The modified Delphi technique, a widely accepted method for the development of consensus amongst experts, was used to generate consensus [20, 21]. As part of this process, a priori decision was made to conduct up to three rounds of anonymous voting or until consensus was achieved for each question,

whichever came first. The survey incorporating the questions and revised recommendations was sent to all panel members. The panel independently and anonymously rated the appropriateness of the recommendations on a five-point Likert scale. Consensus was defined a priori as agreement at $\geq 70\%$ (4 or 5 on the Likert scale) with a minimal response rate of 80%. The results of the survey were tallied and reported to the group. There was 80% survey participation from the members and consensus was achieved on all questions (Table 1).

3 | Results

PICO Question 1

In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis who are not surgical candidates, does airway stenting improve patient-centred outcomes (quality of life, need for repeat procedures) compared to endoscopic interventions?

Recommendations

In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of simple morphology who are not surgical candidates, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures, when frequent endoscopic interventions are required or prior endoscopic interventions have failed. (Conditional Recommendation).

In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of complex morphology who are not surgical candidates, and are not likely to be relieved by endoscopic interventions, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures. (Conditional Recommendation).

Remarks

Patients with simple and complex tracheal stenosis should be discussed in multidisciplinary complex airway teams including but not limited to thoracic surgery, otolaryngology, and interventional pulmonology.

For simple stenoses, endoscopic interventions are the first choice when feasible.

Airway stenting may be considered as a bridge to surgery in select patients initially deemed inoperable, when endoscopic interventions are not feasible or have failed to sustain airway patency.

Literature Review

The approach to a PITS versus PTTS depends on the morphology of the narrowing, and not the underlying aetiology. Simple stenoses, unlike complex stenoses, can be treated with up to 90% success rates with just endoscopic interventions [12, 22]. A study of 167 patients with simple stenoses utilised an approach of

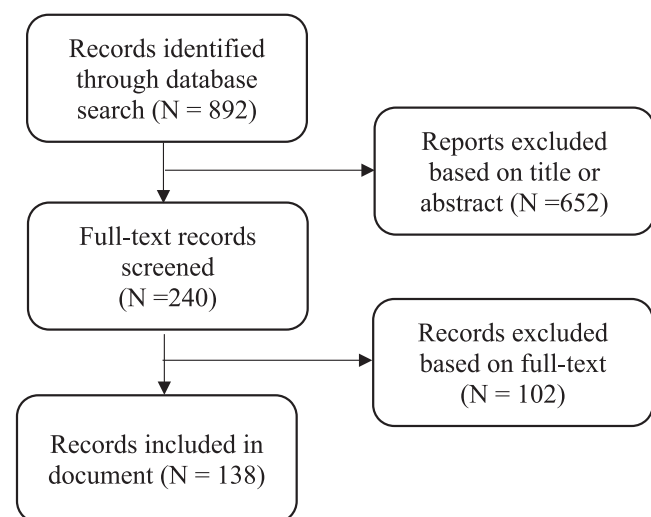


FIGURE 4 | PRISMA diagram.

TABLE 1 | Survey data assessing agreement with recommendation ($n = 16$).

Recommendations	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
PICO Question 1: In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis who are not surgical candidates, does airway stenting improve patient-centred outcomes (quality of life, need for repeat procedures) compared to endoscopic interventions? Recommendations: In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of simple morphology who are not surgical candidates, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures, when frequent endoscopic interventions are required or prior endoscopic interventions have failed. (Conditional Recommendation). In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of complex morphology who are not surgical candidates, and are not likely to be relieved by endoscopic interventions, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures. (Conditional Recommendation).				2 (12.5%)	14 (87.5%)
PICO Question 2: In patients with symptomatic B CAO from fibrotic airway stenosis due to infectious or inflammatory etiologies who are not surgical candidates, does airway stenting improve patient-centred outcomes compared to endoscopic interventions? Recommendations: In patients with symptomatic B CAO from fibrotic airway stenosis caused by infections or inflammatory etiologies, we suggest that airway stenting should be considered to improve patient-centred outcomes only when endoscopic interventions are not feasible or when the stenosis is frequently recurrent. (Conditional Recommendation).				6 (37.5%)	10 (62.5%)
PICO Question 3: In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis does airway stenting improve patient-centred outcomes compared to repeated endoscopic interventions? Recommendations: In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis, we suggest that airway stenting be considered only when endoscopic interventions are not feasible or when the stenosis is recurrent. (Conditional Recommendation).				7 (43.8%)	9 (56.3%)
PICO Question 4: In patients with symptomatic B CAO from excessive central airway collapse does airway stenting improve patient-centred outcomes compared to tracheobronchoplasty? Recommendations: In patients with symptomatic dynamic B CAO from excessive central airway collapse who are surgical candidates, we recommend that airway stenting not be pursued over surgical correction as a definitive therapy. (Strong Recommendation).				2 (12.5%)	14 (87.5%)

(Continues)

TABLE 1 | (Continued)

Recommendations	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
PICO Question 5: In patients with symptomatic B CAO requiring airway stenting, is the use of silicone stents associated with fewer complications compared to fully covered metallic stents? Recommendations: In patients with symptomatic B CAO requiring airway stenting, we recommend silicone stents be used over fully covered metallic stents to minimise long-term stent related complications. (Strong Recommendation).				3 (18.8%)	13 (81.3%)
PICO Question 6: In patients with symptomatic B CAO requiring airway stenting, do pulmonary hygiene measures reduce the incidence of stent-related complications compared to no intervention? Recommendations: In patients with symptomatic B CAO requiring airway stenting, we suggest that pulmonary hygiene measures be implemented in all patients to reduce the incidence of stent-related complications. (Best Practice Statement).				5 (31.3%)	11 (68.8%)
PICO Question 7: In patients with symptomatic B CAO requiring airway stenting, do routine surveillance bronchoscopies reduce the incidence of stent-related complications, compared to only symptom-driven bronchoscopies? Recommendations: In patients with symptomatic B CAO requiring airway stenting, we suggest that routine surveillance bronchoscopies be performed to reduce the incidence of stent-related complications. (Conditional Recommendation).			3 (18.8%)	4 (25%)	9 (56.3%)
PICO Question 8: In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, should stent removal be considered to evaluate for airway remodelling? Recommendations: In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, we suggest that a stent removing trial should be considered at 6–12 months after stenting, to assess for airway remodelling. (Conditional Recommendation).				6 (37.5%)	10 (62.5%)

endoscopic intervention-first and if recurrent, surgery or stenting (in non-operative candidates). The authors reported only a mean of 2 endoscopic procedures per patient, with only 16 and 1 patient requiring stenting and surgery, respectively [12]. In the same study, 42 complex stenoses were managed with either surgery ($N=9$) or, if non-operative candidates, stenting first. The overall success (defined as no recurrence at 1-year follow-up) was 96% and 69% in the simple and complex stenosis groups, respectively.

Most studies do not report separate outcomes of interventions on post-intubation or post-tracheostomy stenoses. Overall, approximately 15%–20% of patients require airway stenting or a definitive intervention such as surgery [12, 23–26]. However, there are no studies comparing surgery to stenting as a first-choice intervention, likely because the standard of care is open resection in surgical candidates. One of the largest series on silicone airway

stenting included 125 stents (5 for simple stenoses and 120 for complex stenoses) [27]. All recurrent simple stenoses were successfully treated and had stents removed, with a mean stent duration of 5.4 months. Among the patients with complex stenoses, only 4 were operable at presentation and 120 were stented. Fourteen of the stent group became operable during the course of follow-up and were treated with surgery. In the remaining 106 patients, no further intervention or additional stenting ('successful treatment') was required in 74 (69.8%). The mean stent duration was 11.9 months in this group.

In the studies that report on subjective improvement [28–30], cough and dyspnoea improvement has been reported in 75%–100% of patients. Forced expiratory volume in 1 s (FEV_1) improvement has been demonstrated in 80% in one 21 patient study [30]. Similar improvements were seen in other small studies [31–34].

Based upon older data, surgical treatment has a failure rate of 5%–15% and a mortality of up to 5% [35, 36]. Complication rates of 15%–40% have been reported, including non-anastomotic complications such as wound infection, bleeding, or glottic dysfunction, and anastomotic complications such as dehiscence, re-stenosis, and granulation [36–39]. When airway stenting is pursued, mucostasis and stent migration are the most commonly reported complications, with incidence rates from 16% to 66% and 2% to 41%, respectively [23–25, 28, 32]. In a study of 128 patients who had silicone stents placed for post-intubation or post-tracheostomy tracheal stenosis, 46 (36%) had their stent removed due to intolerant symptoms, most commonly from stent migration ($N=12$) [40]. Anti-migration stitches or the Montgomery T-tubes (when there is a tracheostomy track) to minimise the stent migration risk have been used in such scenarios [41–43].

Rationale for Recommendations

Therapeutic procedures on stenotic airways can affect patient-centred outcomes by improving the patient's symptoms and quality of life, and avoiding the need for repeated procedures/interventions whilst minimising complications. Studies, however, have largely focused on the latter—technical aspect of interventions in achieving sustained improvement, and not on patient symptoms. We encourage that every case is reviewed by a multi-disciplinary panel. Simple stenoses rarely need surgery or airway stenting, and can be initially managed with endoscopic interventions alone [12]. However, when the morphology of the stenosis is complex, endoscopic interventions are unlikely to succeed. Surgical treatment with a resection and re-anastomosis is the treatment of choice for complex stenoses. When surgery is not feasible, as is often the case [12], airway stenting may be considered as a 'definitive' intervention or a bridge, temporary measure to stabilise the airway until surgery is possible [12, 44]. Even 'definitive' stenting should be viewed as a temporary manoeuvre (discussed in more detail in PICO 8) as the airway could remodel around the stent, and therefore, might not require any subsequent interventions after stent removal. However, stenting is associated with frequent adverse effects including mucostasis, granulation tissue, and migration. An oversized or longer-than-needed stent may injure the airway wall and extend the length of stenosis and render a subsequent surgical intervention not feasible. Appropriate stent sizing and airway hygiene measures (discussed further in PICO 6) are essential to minimise these complications. Given the limited data for anti-migration stitches, the panel cannot give specific recommendations regarding their use.

In certain scenarios wherein airway stenting is not feasible or is technically challenging, performing a tracheostomy to bypass the tracheal stenosis and stabilise the airway is a reasonable option. The use of a Montgomery T-tube can be considered in select scenarios of laryngotracheal stenosis with a tracheostomy tract but should not be needed for most patients with single segment disease alone in the upper or mid-trachea, as is seen with PITS or PTTS. A detailed discussion on the appropriate use of T-tubes or tracheostomy tubes is beyond the scope of this document.

PICO Question 2

In patients with symptomatic B CAO from fibrotic airway stenosis due to infectious or inflammatory etiologies who are not

surgical candidates, does airway stenting improve patient-centred outcomes compared to endoscopic interventions?

Recommendations

In patients with symptomatic B CAO from fibrotic airway stenosis caused by infections or inflammatory etiologies, we suggest that airway stenting should be considered to improve patient-centred outcomes only when endoscopic interventions are not feasible or when the stenosis is frequently recurrent. (Conditional Recommendation).

Remarks

The above recommendation refers to fibrotic airway stenosis. During the active/acute inflammatory phase of diseases, every attempt must be made to avoid airway stenting, unless there is no other option available and a critical stenosis needs to be managed urgently in a symptomatic patient.

Endoscopic interventions, when feasible, should be attempted first for both post-infectious and post-inflammatory stenoses.

Surgical resection should be very carefully considered in patients with stenosis due to GPA, given high post-operative recurrence rates and the often multifocal airway involvement.

Literature Review

In a 75-patient study of post-tuberculosis airway stenosis, dyspnoea was relieved in 88% post-stenting and the median improvement in FEV_1 was 22% [45]. Larger retrospective series of over 300 patients have evaluated the utility of silicone stenting in B CAO from post-tuberculosis airway stenosis [46, 47]. However, these studies do not report on patient symptoms or quality of life after the intervention. While granulation tissue and mucostasis were more common with Y stents than straight ones (64% vs. 17% and 18% vs. 2%, respectively), migration (11%) and re-stenosis (19%) were commonly seen with the latter. Around 70% of these stents were removed after a median of 25 months, with no need for further interventions. Younger age, male sex, absence of parenchymal calcifications or lobar or segmental consolidation, and a stent duration of more than 12 months were factors related to successful stent removal [47]. However, results for successful stent removal after suspected airway remodelling have not been as encouraging in smaller series. These studies also report high complication rates warranting stent removal [44, 48–50].

A two-centre retrospective study evaluated the utility of uncovered SEMS in 27 patients with B CAO from relapsing polychondritis [51]. Improvements in FEV_1 , peak expiratory flow, 6-min walk distance, and mMRC dyspnoea scores were noted. The small size, retrospective design, and poor longitudinal follow-up limit the generalisability of the findings of this study; and caution must be exercised when considering SEMS (especially uncovered) in any B CAO aetiology. In a study of 47 patients with GPA (36 with subglottic stenosis and 11 with bronchial stenosis), failure of the first endoscopic treatment was reported in 49% at 1 year, 70% at 2 years, and 80% at 5 years [52]. A shorter time from

diagnosis to procedure and bronchial stenosis was associated with a higher incidence of treatment failure, while prednisone > 30 mg/day was associated with a lower incidence of failure. Nine patients with subglottic stenoses and 10 with bronchial stenosis needed stents; however, outcomes of and follow-up after stenting were not reported.

Rationale for Recommendations

Data for airway stenting in B CAO is largely limited to patients with fibrotic stenoses after tuberculosis. There are no dedicated case series of airways stenting for patients with other common inflammatory etiologies of B CAO such as inflammatory bowel disease, GPA, and sarcoidosis. However, management principles for these conditions may be extrapolated from the post-tuberculosis B CAO literature, with the exception of small studies that show that recurrence warranting subsequent procedures tends to be high in patients with GPA who undergo open surgical resection [53, 54]. In general, as with PITS and PTTS, an endoscopic intervention should be attempted first if the morphology of the stenosis is simple. The threshold to intervene, and especially to stent, when there is active disease should be much higher, barring extraneous circumstances with critical airway narrowing not amenable to simple interventions alone. Airway stenting should largely be reserved for fibrotic post-inflammatory stenoses. When pursued, silicone stenting should be offered, and the patients should be followed up closely to monitor for complications and evaluate for stent removal after 6–12 months (see PICO 8). Unlike PITS and PTTS, post-inflammatory or post-infectious stenoses are often bronchial, or concurrent tracheal and bronchial, and thus, unamenable to curative-intent surgery. Complex or recurrent short segment tracheal or bronchial stenoses should be approached in a multidisciplinary fashion to discuss the role of surgery as a definitive alternative to airway stenting when the acute inflammatory phase of the underlying disease has resolved. This is more likely the case with post-infectious etiologies as autoimmune disorders have a high recurrence rate at 5 years [52]. Disease relapse, anastomotic complications, and the need for immunosuppressive drugs that impair healing make surgery an unfavourable choice for recurrent stenosis in patients with autoimmune diseases such as GPA [53]; however, it still remains an option in highly selected individuals [54].

Bronchial stenosis can occur in up to 4%–25% of patients that received 74 Gy to 86 Gy external beam radiation [55] and in up to a third of patients with centrally located tumours treated with stereotactic body radiation therapy [56]. Moreover, stenosis is often a late complication of radiation, often manifesting up to 2 or more years after treatment [57]. Our recommendations and remarks are largely applicable to post-radiation stenoses.

3D printed silicone stents may allow for better conformation to complex, multi-segmental airway stenoses and, when available, may be an option for certain airways that are not amenable to traditional stenting techniques. Case reports of the use of 3D printed stents for benign stenoses have been published [58], but reliable data demonstrating greater adaptability or a better safety profile of these stents are yet to be published.

PICO Question 3

In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis, does airway stenting improve patient-centred outcomes compared to repeated endoscopic interventions?

Recommendations

In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis, we suggest that airway stenting be considered only when endoscopic interventions are not feasible or when the stenosis is recurrent. (Conditional Recommendation).

Remarks

Endoscopic interventions, when feasible, should be attempted first.

Literature Review

In a study of 223 lung transplantation patients with 345 anastomoses, 70 anastomoses (20.2%) had complications needing interventions [17]. Eventually, 47 (21%) patients underwent SEMS airway stenting. Amongst these, an FEV₁ improvement of 370 mL was noted ($p < 0.005$). The most common complications reported were granulation tissue (57%), mucus (6.3%) and migration (8.6%). Similar results were obtained in another 24-patient study [59]. In a study in which 23 silicone stents were placed in 17 transplant patients, symptomatic improvement was noted in all and the mean FEV₁ improvement was 672 mL [60]. Similar to the above studies [17, 59], granulation tissue (43%), mucus plugging (30%) and migration (30%) were commonly noted. Most stent-related complications were managed bronchoscopically; however, in a small study, 25% of stents were removed due to stent-related granulation tissue or mucus plugging [61]. Uncovered SEMS have also been used in lung transplantation patients with close follow-up, as was reported in a study of 111 SEMS (91% of which were uncovered) in 65 patients [62]. Clinical improvement was noted in 80% and the mean FEV₁ improvement was 299 mL (21%). However, stent-related complications were noted in 52 (80%), with re-stenosis being the most frequent, noted in 34 (52%) patients.

A large 30-year review of 651 patients with lung transplantations reported 76 anastomotic complications in 9.6% of patients (6.4% per anastomosis), of which the majority were stenosis (67%) and malacia (15%) [63]. Forty-two (55%) needed airway stenting. While the overall median survival was shorter in those who had anastomotic complications (101 vs. 136 months; $p = 0.044$), amongst those with anastomotic complications, there was no difference in survival in those who needed stenting versus those who did not (102 vs. 91 months; $p = 0.627$). In another study in which 23 patients (56%) received airway stents due to recurrent stenosis or balloon dilation failure, the mean survival in patients with stents was longer than in those who underwent dilations alone (82 vs. 22 months, respectively), and stent insertion was associated with a 66% reduction in the risk of death ($p < 0.02$) [64].

In a single centre study that reviewed 1170 lung transplants and included 135 patients with post-transplant B CAO, airway

stenting was considered after more than two endoscopic interventions or when significant malacia was present [65]. One hundred and fourteen biodegradable stents were inserted in 78 patients. Migration (2%), secretions requiring bronchoscopy (9%), granulation tissue needing endoscopic recanalization (13%) and re-stenosis after stent degradation (18%) were complications noted. Fourteen patients underwent surgery (including 6 re-transplantations) due to complications from their B CAO. Small series have also reported the use of 3D printed stents for complex, post-lung transplant anastomotic B CAO [66, 67].

Rationale for Recommendations

Anastomotic complications are relatively common after lung transplantation, with an incidence of 6%–20% per anastomosis [17, 62, 63, 68, 69]. Complications are frequently encountered with airway stenting for any benign aetiology of CAO, and lung transplant airways with anastomotic B CAO are no different. Our recommendations are limited to B CAO, which includes stenosis and/or malacia post-lung transplantation, and should not be extrapolated to stenting for anastomotic dehiscence. Re-operation or transplantation is rarely an option for anastomotic bronchial CAO. When surgery is feasible, any airway stenting consideration should be discussed in a multidisciplinary forum. In general, when endoscopic interventions fail or are frequently required to maintain airway patency, airway stenting must be considered. Due to the hope of airway remodelling around the stent, these stents are usually temporary, which has generated interest in the use of biodegradable stents [65]. Moreover, anastomotic bronchial stenoses are anecdotally often of complex morphology, and often concurrently affect mainstem bronchi and lobar airways. This has generated interest in 3D printed stents for this indication [66]. Due to the paucity of published data and the lack of widespread availability, biodegradable or 3D printed stent use cannot be routinely recommended at this time.

While the data and our recommendations are pertinent to post-transplantation anastomotic complications, in select scenarios they may be applicable to some patients with other post-surgical B CAO such as after sleeve resection. These cases must be approached on a case-by-case basis, in a multidisciplinary forum.

PICO Question 4

In patients with symptomatic B CAO from excessive central airway collapse, does airway stenting improve patient-centred outcomes compared to tracheobronchoplasty?

Recommendations

In patients with symptomatic dynamic B CAO from excessive central airway collapse who are surgical candidates, we recommend that airway stenting not be pursued over surgical correction as a definitive therapy. (Strong Recommendation).

Remarks

Tracheobronchomalacia and excessive dynamic airway collapse are distinct entities, and most studies reporting on stenting or tracheobronchoplasty (TBP) have addressed EDAC, not TBM.

We suggest management of comorbidities and the use of as needed non-invasive positive pressure ventilation as the first line of treatment. Only when these measures fail, in patients who have clearly attributable symptoms from excessive central airway collapse (ECAC), should surgery be considered. Stent trials prior to TBP must be considered on a case-by-case basis.

We suggest that airway stenting as a long-term therapy for ECAC should only be an absolute last-resort intervention, as there are associated frequent stent-related complications. Stenting as a definitive therapy may only be attempted in those who are not surgical candidates or have failed surgery (and are not re-operative candidates) and have failed exhaustive co-morbidity management and non-invasive positive pressure ventilation trials.

Literature Review

Short term stenting for airway stabilisation in patients with ECAC can improve symptoms and quality of life [70]. Stent trials have been used to predict response to airway stabilisation with tracheobronchoplasty (TBP) [71–73], which in experienced hands, in well-selected patients, has been shown to improve symptoms, quality of life and functional status [74, 75]. In some series, patients have benefited from TBP without prior stenting trial [76]. However, our PICO question relates to stents as a destination therapy. A small study that utilised silicone Y stents for 7 patients with excessive dynamic airway collapse (EDAC) with follow-up noted no significant change in FEV₁ and a mean of 2.2 follow-up bronchoscopies per patient due to frequent complications (mucus plugging in 6, granulation tissue in 3, stent migration in 1 and death due to unclear reasons in 1) [77]. Silicone stents were used to treat 58 patients with severe ‘tracheobronchomalacia (TBM)’ (however, the study likely included patients with both TBM and EDAC) and resulted in dyspnoea and/or secretion clearance improvement in 45 (77.5%), but no improvement in FEV₁ [70]. Complications however were frequent (90%) and included mucus plugging (N = 21), infection (N = 14), migration (N = 10), severe cough (N = 3), subglottic edema (N = 3) and stent breakage (N = 1).

Rationale for Recommendations

While TBM and EDAC are terms that have been used interchangeably under the umbrella of ECAC, the panel considers them as distinct conditions, with the former representing a primary airway pathology and the latter usually just being an epiphenomenon of high peripheral airway resistance or increased pleural pressure resulting in collapse of the central airways at their weakest point, as can be seen in patients with asthma, COPD, and morbid obesity [78]. The presence of EDAC (often up to 80%–90% cross-sectional area reduction) in healthy volunteers [79] and the reports of resolution of severe EDAC in patients with COPD after lung transplantation [80] or after treating COPD exacerbation [81] support the notion of EDAC being a secondary phenomenon rather than a primary central airway pathology.

Dyspnoea, cough, and secretion management in patients in TBM or EDAC can be achieved with the application of as needed non-invasive positive airway pressure (with the use of continuous or bi-level positive airway pressure) to pneumatically stent open the airway [82–84]. Optimal management of underlying comorbid conditions such as COPD or asthma, obesity, or

gastroesophageal reflux disease (associated with ECAC), nocturnal and as-needed positive airway pressure, and mucociliary clearance techniques such as flutter valves and mucolytics must be considered or trialled prior to committing a patient to an airway stent [85]. Airway stenting as a trial to identify patients who may benefit from surgery [73] has become an acceptable practice, although surgical series describe positive outcomes without the need for a stenting trial [76]. However, airway stenting as a long-term therapy is associated with a very high complication rate and cannot be considered as an alternative to surgery. Even in medically inoperable patients, the threshold to insert a stent must be very high (all other treatment pathways must have failed and any alternative explanations for dyspnoea must be excluded). Potential surgical candidates must be evaluated and managed at a centre with expertise in surgery for TBM.

Despite the lack of high certainty of evidence in this area, the panel feels that due to the high risk of complications associated with destination airways stenting for ECAC, a strong recommendation is warranted considering the larger potential for benefit with surgical correction, when feasible.

PICO Question 5

In patients with symptomatic B CAO requiring airway stenting, is the use of silicone stents associated with fewer complications compared to fully covered metallic stents?

Recommendations

In patients with symptomatic B CAO requiring airway stenting, we recommend silicone stents be used over fully covered metallic stents to minimise long-term stent related complications. (Strong Recommendation).

Remarks

When expertise in rigid bronchoscopy is not available, in non-emergent scenarios, we suggest patient transfer to a facility with such expertise. When rigid bronchoscopy is not feasible due to anatomical factors, fully covered metallic stenting may be considered.

There is no circumstance in which an uncovered metallic stent should be used as a destination therapy for B CAO.

Literature Review

Complications with airway stents, especially when in situ for months to years, are common. However, complications such as mucus plugging and migration often occur soon after stent placement [86]. While there are no comparative prospective studies evaluating the complications with silicone versus metallic stents for B CAO, studies reporting the use of metal stents for B CAO are few. Retrospective series report similar granulation tissue and migration rates for silicone versus metallic stents—23%–53% and 1%–48% versus 7%–57%, and 14%–32%, respectively [17, 23, 25, 28, 32, 40, 59, 86–88]. Mucus plugging is reported in 28%–66% of patients who undergo silicone stenting [23, 25, 28, 32, 40, 86–88]. Consequently, stent-associated respiratory tract infections may occur (in up to 19% patients in older series) which are associated with higher mortality rates [89].

Rationale for Recommendations

In the absence of comparative studies and given the paucity of data on metallic airway stenting outside the setting of lung transplant anastomotic complications, no definitive conclusions can be made on whether silicone or metallic stenting is better for complication mitigation in patients with B CAO. Retrospective studies, reporting biases, and varied definitions may account for the large variation in the reported incidence of these complications, as is also seen in the airways stenting in malignant CAO literature [90]. Stents placed for B CAO are expected to be left in situ for at least 6–12 months (see PICO 8). Prolonged use of metallic stents is associated with tissue ingrowth (through uncovered portions) and fractures (Figure 5) that consequently make stent removal challenging; these complications are not seen with silicone stents.

In the absence of conclusive evidence, the panel is unable to provide firm evidence-based guidance on the best stent type from the existing literature. However, given the long record of efficacy, manageable complications, and ease of removability of silicone stenting [91, 92], the panel recommends their use over metallic stents for B CAO, when technically feasible. Moreover, benign stenoses are often of complex morphology, and silicone stents can be easily manipulated or customised on site. When expertise in rigid bronchoscopy is not available, we suggest patient transfer to a facility where such expertise is available for consideration for silicone airway stenting. Attractive features of fully covered metallic stenting include the ability to insert them using flexible bronchoscopy and a more favourable internal to external diameter. Consistent with the 2005 FDA advisory (<http://www.fda.gov/MedicalDevices/Safety/AlertsandNotices/PublicHealthNotifications/ucm062115.htm>) we recommend against uncovered metallic stenting for any aetiology of B CAO. There are mechanical differences between fully covered SEMS based on the structure of the metal mesh. SEMS manufactured as a monofilament may have greater resistance to mechanical manoeuvres, such as twisting or pulling, compared to those fabricated from separate metal rings.

Despite the lack of high certainty of evidence in this area, the panel feels that due to the large body of evidence and experience favouring silicone stenting for B CAO, a strong recommendation is warranted considering the larger potential for benefit and apparent absence of harm.

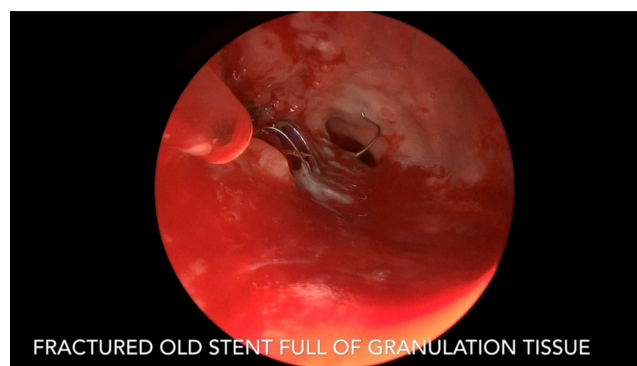


FIGURE 5 | An embedded uncovered metallic stent which has fractured, with struts occluding the tracheal lumen, noted during rigid bronchoscopy.

PICO Question 6

In patients with symptomatic B CAO requiring airway stenting, do pulmonary hygiene measures reduce the incidence of stent-related complications compared to no intervention?

Recommendations

In patients with symptomatic B CAO requiring airway stenting, we suggest that pulmonary hygiene measures be implemented in all patients to reduce the incidence of stent-related complications. (Best Practice Statement).

Remarks

There is no evidence favouring a specific pulmonary hygiene regimen or saline nebulization concentration.

Literature Review

There are no studies comparing one pulmonary hygiene regimen (including expectorants, mucolytics and mechanical techniques) to another. A recent pilot, randomised study demonstrated that the use of three times a day 3% saline nebulization compared to normal saline may lead to fewer unscheduled bronchoscopies due to symptoms from stent-related mucus plugging [93].

Rationale for Recommendations

All members on this panel use pulmonary hygiene measures for patients with airway stents placed for B CAO. We acknowledge that it is difficult to predict which patients are at high risk for significant stent-related mucus plugging and that some pulmonary hygiene measures are expensive. Published evidence shows significant practice variation in regimens used across institutions, with normal saline being the most commonly used agent [94]. Expectorants, mucolytics, and mechanical techniques such as flutter valves are also commonly used to assist with secretion mobilisation and clearance in patients with indwelling airway stents. Based on anecdotal evidence of higher rates of biofilm formation and mucus plugging in patients with airway stents who are non-compliant with the pulmonary hygiene regimens, the panel suggests that these measures (involving saline nebulization with or without mucolytics or expectorants) be implemented in all patients after airway stenting for B CAO.

PICO Question 7

In patients with symptomatic B CAO requiring airway stenting, do routine surveillance bronchoscopies reduce the incidence of stent-related complications compared to only symptom-driven bronchoscopies?

Recommendations

In patients with symptomatic B CAO requiring airway stenting, we suggest that routine surveillance bronchoscopies be performed to reduce the incidence of stent-related complications. (Conditional Recommendation).

Remarks

The optimal surveillance frequency remains unclear. We suggest that the first surveillance bronchoscopy be performed in 4–6 weeks after airway stenting. The need and timing for further scheduled surveillance bronchoscopies must be determined by the provider factoring in the risk of stent-related complications, expected duration of the indwelling stenting, and the risks associated with subjecting the patient to subsequent bronchoscopies.

Surveillance bronchoscopies do not preclude the need for clinical follow-up. Chest computed tomography may complement, but not replace bronchoscopic surveillance.

Literature Review

A single centre, retrospective study of 70 patients reviewed surveillance ($N=39$; within 2–3 months) versus emergent-only ($N=31$) bronchoscopies in patients undergoing airway stenting (57% with a benign aetiology) [95]. In the routine surveillance group, amongst 31 asymptomatic patients, complications were detected in 9. Overall, symptoms seemed to be predictive of complications in 33 (85%) of the 39 symptomatic patients (8 in the surveillance group and 31 in the emergent-only group). While this may suggest that bronchoscopies triggered by symptoms may suffice for most, overall, 44% of surviving patients needed an emergent bronchoscopy within 2–3 months, possibly suggesting that a surveillance procedure may be warranted sooner than 2 months. A single centre, retrospective study of 134 patients (36% with a benign etiologies) reported outcomes associated with surveillance bronchoscopies performed 4–6 weeks after airway stenting in 94 patients (100 stents) [96]. Stent-related complications were detected in 69%; there was no difference in complications on a univariate analysis for M CAO versus B CAO (OR 0.93 (0.4–2.19)). Hybrid stents were more commonly associated with migration and mucus plugging. The stent indication, its location, and the presence or absence of symptoms were not predictive of detecting a complication at the surveillance bronchoscopy ($p=0.15$). Moreover, 60% of asymptomatic patients had at least one stent-related complication noted. Another study with a more intense surveillance regimen, without a comparison group, suggested that complications were detected more often in patients with SEMS implanted for B CAO than M CAO (42.2% v 21.1%, $p=0.001$) [97].

Rationale for Recommendations

After more than three decades of dedicated airway stenting, the most optimal surveillance protocol after airway stent insertion remains uncertain and institutional practices vary [98]. Surveillance, whether clinical, with chest CT or bronchoscopy, aims to detect complications related to airway stents earlier, at a time when an intervention may be easier and thereby avoid symptoms, urgent procedures, or emergency room visits. Moreover, emergent procedures, as can be seen after a stent migrates or due to severe stent-related mucostasis or granulation tissue, may be associated with more procedure-related complications [99]. Bronchial washings for microbiological culture can also be obtained during bronchoscopies, which may help the clinician to prescribe an appropriate antibiotic when clinically indicated. Moreover, small granulomas can be removed during

a bronchoscopy using a forceps or by cryoextraction. The limited published evidence suggests that timely detection of stent-related problems may be achieved with a surveillance bronchoscopy performed within 2 months after the stent insertion. In general, bronchoscopy should not absolve the need for clinical follow-up. Chest CT, while a useful adjunct [100], has a low sensitivity and specificity to detect endobronchial pathologies [101], and should not be utilised as a stand-alone surveillance tool.

PICO Question 8

In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, should stent removal be considered to evaluate airway remodelling?

Recommendations

In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, we suggest that a stent removal trial should be considered at 6–12 months after stenting, to assess for airway remodelling. (Conditional Recommendation).

Remarks

Providers must factor in the likelihood of airway remodelling based on the cartilage integrity on imaging, underlying aetiology, and the patient's tolerance of the indwelling stent, when determining when and whether the stent removal trial should be performed.

Providers must be mindful that stent removal is a trial, and there is a significant risk of re-stenosis requiring re-intervention within days or weeks post stent removal.

Literature Review

A few studies have reported successful stent removal after it was initially placed for B CAO. Successful removal without the need for subsequent intervention ('airway remodelling') appears to occur in 30%–40% patients after a stent indwelling period of at least 6–12 months (or longer) [23, 40, 44, 102–105]. In the largest study that included 128 patients who had silicone stents placed for PITS or PTTS, 46 (36%) had their stent removed due to intolerant symptoms, most commonly from migration (in 12 patients) [40]. Overall, stents were removed at 4–6 months, and at 1 year follow up post-removal the airway remained stable (aka remodelled) in 54 (42%) patients. The remaining 74 (58%) patients had tracheal restenosis that occurred at a median of 1 week after stent removal. Diabetes and morbid obesity were independent predictors of restenosis.

Rationale for Recommendations

Airway stenting for B CAO, if tolerated, may lead to airway stability (potentially due to remodelling) in up to 40% of patients after 6–12 months of having the stent in place. A few experts on this panel perform the stent removal trial in 12–18 months. A stent removal trial to assess for stability could be considered at this timeframe. Remodelling likely occurs from a regeneration sequence involving inflammation, followed by epithelial and

cartilage formation [106, 107]. Logically, the remodelling effect may be more likely to occur if the stent was placed for fibrotic stenosis without extensive cartilage damage, but the evidence separating outcomes based on this hypothesis is lacking.

When tracheal stent removal trials are being performed, the panel suggests that patients be observed in-house post-intervention for at least 12–24 h to monitor the need for an emergent intervention due to early disease relapse. In general, if the airway stenosis recurs, it usually becomes apparent within days or weeks, so patients should be closely followed after stent removal. While we advocate for silicone stenting for B CAO, there are circumstances in which a SEMS might have been placed (e.g., a patient in extremis when rigid bronchoscopy is unavailable or not feasible due to anatomical reasons). SEMS removal may be associated with more complications, especially if the stents are embedded. A 2010 study of removal of 55 SEMS in 46 patients (80% with B CAO) reported that patients required hospitalisation and ICU admission in 78% and 39% of the encounters, with a median length of stay of 3.5 and 0 days, respectively [92]. However, these stents were in situ for an average of 292 days. Shorter-term SEMS, however, are not expected to have such high rates of adverse events [91, 108].

4 | Conclusions and Future Directions

This guideline document intends to offer guidance regarding airway stenting in patients with symptomatic B CAO from various commonly encountered aetiologies (Table 2). Recommendations are based on standard consensus methodology after an extensive literature review and opinions from 20 experts from 19 institutions across 12 countries. The suboptimal quality of existing evidence, however, limits the strength of the recommendations. Data is largely retrospective and single-centred. Ethical considerations, acuity of presentation, and variability in type, location, severity, and vertical extent of B CAO, as well as lack of uniformly accepted classification or scoring systems, understandably, make it challenging for investigators to recruit for randomised, prospective studies.

As surgical and endoscopic techniques continue to improve, the threshold for airway stenting may be higher in the future. On the other hand, stent design improvements may lead to lower complications. Patient-specific 3D stents designed and prepared using the patient's CT scan or an airway model have been used for complex benign stenoses [66, 109–112]. Large scale studies on 3D printed stents however, are yet to be conducted. Nevertheless, their theoretical appeal comes from a better fit conforming to the patient's airway anatomy, but data on granulation tissue and migration rates of these stents remain to be published. Drug-eluting stents [113] or the use of novel biocompatible materials with antimicrobial properties [114], in principle, may reduce biofilm formation or granulation, but in-human testing has not yet been conducted. Biodegradable stents [115, 116] have been used in small studies involving patients with lung transplant anastomotic complications [63] and growing-paediatric airways with severe malacia [117]. Their properties, including time to degradation and complication profile in different disease states, still need to be studied before their use in B CAO can be recommended.

TABLE 2 | PICO questions and recommendations.

PICO Question 1: In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis who are not surgical candidates, does airway stenting improve patient-centred outcomes (quality of life, need for repeat procedures) compared to endoscopic interventions?

Recommendations:

- **In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of simple morphology who are not surgical candidates, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures, when frequent endoscopic interventions are required or prior endoscopic interventions have failed. (Conditional Recommendation).**
- **In patients with symptomatic B CAO from post-intubation or post-tracheostomy tracheal stenosis of complex morphology who are not surgical candidates, and are not likely to be relieved by endoscopic interventions, we suggest that airway stenting be pursued to improve patient-centred outcomes such as quality of life and reduce the need for recurrent procedures. (Conditional Recommendation).**

Remarks:

- *Patients with simple and complex tracheal stenosis should be discussed in multidisciplinary complex airway teams including but not limited to thoracic surgery, otolaryngology and interventional pulmonology.*
- *For simple stenoses, endoscopic interventions are the first choice, when feasible.*
- *Airway stenting may be considered as a bridge to surgery in select patients initially deemed inoperable, when endoscopic interventions are not feasible or have failed to sustain airway patency.*

PICO Question 2: In patients with symptomatic B CAO from fibrotic airway stenosis due to infectious or inflammatory etiologies who are not surgical candidates, does airway stenting improve patient-centred outcomes compared to endoscopic interventions?

Recommendations:

- **In patients with symptomatic B CAO from fibrotic airway stenosis caused by infections or inflammatory etiologies, we suggest that airway stenting should be considered to improve patient-centred outcomes only when endoscopic interventions are not feasible or when the stenosis is frequently recurrent. (Conditional Recommendation).**

Remarks:

- *The above recommendation refers to fibrotic airway stenosis. During the active/acute inflammatory phase of diseases, every attempt must be made to avoid airway stenting, unless there is no other option available and a critical stenosis needs to be managed urgently in a symptomatic patient.*
- *Endoscopic interventions, when feasible, should be attempted first for both post-infectious and post-inflammatory stenoses.*
- *Surgical resection should be very carefully considered in patients with stenosis due to GPA, given high post-operative recurrence rates and the often multifocal airway involvement.*

PICO Question 3: In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis does airway stenting improve patient-centred outcomes compared to repeated endoscopic interventions?

Recommendations:

- **In patients with symptomatic bronchial B CAO from post-lung transplant airway stenosis, we suggest that airway stenting be considered only when endoscopic interventions are not feasible or when the stenosis is recurrent. (Conditional Recommendation).**

Remarks:

- *Endoscopic interventions, when feasible, should be attempted first.*

PICO Question 4: In patients with symptomatic B CAO from excessive central airway collapse does airway stenting improve patient-centred outcomes compared to tracheobronchoplasty?

Recommendations:

- **In patients with symptomatic dynamic B CAO from excessive central airway collapse who are surgical candidates, we recommend that airway stenting not be pursued over surgical correction as a definitive therapy. (Strong Recommendation).**

Remarks:

- *Tracheobronchomalacia and excessive dynamic airway collapse are distinct entities and most studies reporting on stenting or tracheobronchoplasty (TBP) have addressed EDAC, not TBM.*
- *We suggest management of comorbidities and the use of as needed non-invasive positive pressure ventilation as the first line of treatment. Only when these measures fail, in patients who have clearly attributable symptoms from excessive central airway collapse (ECAC), should surgery be considered. Stent trials prior to TBP must be considered on a case-by-case basis.*
- *We suggest that airway stenting as a long-term therapy for ECAC should only be an absolute last-resort intervention, as there are associated frequent stent-related complications. Stenting as a definitive therapy may only be attempted in those who are not surgical candidates or have failed surgery (and are not re-operative candidates) and failed exhaustive co-morbidity management and non-invasive positive pressure ventilation trials.*

PICO Question 5: In patients with symptomatic B CAO requiring airway stenting, is the use of silicone stents associated with fewer complications compared to fully covered metallic stents?

Recommendations:

- **In patients with symptomatic B CAO requiring airway stenting, we recommend silicone stents be used over fully covered metallic stents to minimise long-term stent related complications. (Strong Recommendation).**

Remarks:

- *When expertise in rigid bronchoscopy is not available, in non-emergent scenarios, we suggest patient transfer to a facility with such expertise. When rigid bronchoscopy is not feasible due to anatomical factors, fully covered metallic stenting may be considered.*
- *There is no circumstance in which an uncovered metallic stent should be used as a destination therapy for B CAO.*

(Continues)

TABLE 2 | (Continued)

PICO Question 6: In patients with symptomatic B CAO requiring airway stenting, do pulmonary hygiene measures reduce the incidence of stent-related complications compared to no intervention?	
Recommendations:	
In patients with symptomatic B CAO requiring airway stenting, we suggest that pulmonary hygiene measures be implemented in all patients to reduce the incidence of stent-related complications. (Best Practice Statement).	
Remarks:	
There is no evidence favouring a specific pulmonary hygiene regimen or saline nebulization concentration.	
PICO Question 7: In patients with symptomatic B CAO requiring airway stenting, do routine surveillance bronchoscopies reduce the incidence of stent-related complications, compared to only symptom-driven bronchoscopies?	
Recommendations:	
In patients with symptomatic B CAO requiring airway stenting, we suggest that routine surveillance bronchoscopies be performed to reduce the incidence of stent-related complications. (Conditional Recommendation).	
Remarks:	
- The optimal surveillance frequency remains unclear. We suggest that the first surveillance bronchoscopy be performed in 4–6 weeks after airway stenting. The need and timing for further scheduled surveillance bronchoscopies must be determined by the provider factoring in the risk of stent-related complications, expected duration of the indwelling stenting and the risks associated with subjecting the patient to subsequent bronchoscopies. - Surveillance bronchoscopies do not preclude the need for clinical follow-up. Chest computed tomography may complement, but not replace bronchoscopic surveillance.	
PICO Question 8: In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, should stent removal be considered to evaluate for airway remodelling?	
Recommendations:	
In patients with symptomatic B CAO from airway stenosis of any aetiology requiring airway stenting, we suggest that a stent removing trial should be considered at 6–12 months after stenting, to assess for airway remodelling. (Conditional Recommendation).	
Remarks:	
- Providers must factor in the likelihood of airway remodelling based on the cartilage integrity on imaging, underlying aetiology, and the patient's tolerance of the indwelling stent, when determining when and whether the stent removal trial should be performed. - Providers must be mindful that stent removal is a trial, and there is a significant risk of re-stenosis requiring re-intervention within days or weeks post stent removal.	

Given that disease specific procedural volumes of B CAO at single centres are unlikely to be robust, future studies should seek multicentre collaboration. Future studies, including those with newer stent designs, must continue to focus on intermediate and long-term patient-centred outcomes, and not just technical success and immediate symptomatic improvement. We anticipate that as further evidence is generated, future iterations of these guidelines will provide supplemental or different recommendations.

Conflicts of Interest

H.D. serves as a consultant for Novatech. M.O. has received speaker fees from Olympus. S.M. serves as an educational consultant for Olympus, Johnson & Johnson, and Boston Scientific. The other authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.